

3 Organisms can be classified using the three-domain system.

Molecular evidence is used to place an organism into one of the three domains.

- (a) (i) Give **one** example of the molecular evidence used to support the three-domain system.

(1)

- (ii) Describe the role of the scientific community in evaluating the evidence for this system of classification.

(2)

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(b) The table shows some features of living organisms.

Which box in each row shows whether the feature is found in these domains?

(4)

Feature	Archaea only	Archaea and Bacteria only	Archaea and Eukarya only	Archaea and Bacteria and Eukarya
absence of a nuclear envelope	☐	☐	☐	☐
presence of circular DNA	☐	☐	☐	☐
presence of a cell membrane	☐	☐	☐	☐
presence of ribosomes	☐	☐	☐	☐

(Total for Question 3 = 7 marks)

